

MOHAMMED SHADEED P

B.Tech CSE (Data Science) Student | Data Analytics | Aspiring Data Scientist

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EDUCATION

Mar Athanasius College of Engineering, Kothamangalam

2023 – 2027

B.Tech – Computer Science & Engineering (Data Science)

CGPA: 6.86 (up to Semester 6) | Currently in Semester 7 | Expected Graduation: 2027

GVHSS Arimbra

2022

Higher Secondary Education (HSE) – 88.6%

VHM Higher Secondary School, Morayur

2020

Secondary School Leaving Certificate (SSLC) – 94.4%

SKILLS

Languages: C, C++, Java, JavaScript, Python

Frameworks & Libraries: React.js, Node.js, Express.js, Flask, FastAPI, Pandas

AI / Data: Machine Learning, Generative AI, AI Strategy, Scikit-learn, Data Analytics

Tools & Platforms: Arduino IDE, Git, VS Code, MySQL Workbench, Postman

Databases: MySQL, MongoDB, SQLite

Hardware & Embedded: ESP32, MPU-6050 (IMU), SIM800L (GSM), NEO-6M (GPS), I2C / UART protocols

Relevant Coursework: OOP, DBMS, Machine Learning, Data Science, Software Engineering, Computer Networks, Operating Systems, Probability & Statistics

PROJECTS

WaveGuard – IoT Coastal Early Warning System

ESP32, MPU-6050, SIM800L (GSM), Python, FastAPI, React.js, SQLite, Open-Meteo API, Arduino IDE

- Developed a buoy-based IoT monitoring system to detect Kallakkadal (dangerous fair-weather coastal swell surges) along the Kanayannur coastline, Kerala, enabling early warnings for fishing communities.
- Built ESP32 firmware to sample MPU-6050 IMU g-force across 20-sample windows, classify sea conditions into CALM / WATCH / WARNING states, and transmit readings every 5 seconds via HTTP.
- Integrated the Open-Meteo satellite wave API with a custom Python fusion engine to cross-validate buoy sensor data with real-time swell energy metrics for improved alert reliability.
- Implemented a bilingual (English & Malayalam) React dashboard with a JWT-authenticated admin console, and GSM SMS alerts via SIM800L with a 1000µF bulk capacitor handling 2A transmission spikes.

Smart Tourism Recommendation System

Python (Flask), React.js, Pandas, Scikit-learn (K-Means, Apriori)

- Developed a personalized travel recommendation engine using K-Means clustering to segment users into budget and preference profiles, enabling targeted itinerary suggestions.
- Implemented the Apriori algorithm to mine frequent travel pattern associations and generate customized multi-destination itineraries with cost estimates.
- Built a full-stack web interface with a Flask REST backend and React.js frontend, integrating data-processing pipelines using Pandas for user profile analysis.

Smart Waste Management System

MySQL, PHP, HTML, CSS, JavaScript

- Built a web platform to monitor real-time waste bin fill levels and optimize collection routes for sanitation drivers and administrators.
- Designed and implemented an Admin Dashboard with full CRUD capabilities to manage user complaints, driver assignments, and live bin status updates.
- Structured a normalized relational database schema in MySQL to efficiently track bin sensors, driver records, and complaint lifecycle management.

CERTIFICATIONS

- Programming in Java – NPTEL, IIT Kharagpur: 12-week certification (Ministry of Education, Govt. of India funded); Elite status, consolidated score 69% (25/25 assignments, 43.75/75 proctored exam).
- GenAI Powered Data Analytics Job Simulation – Tata
- Data Analytics Job Simulation – Deloitte Australia

ACHIEVEMENTS & ACTIVITIES

- Designed and developed WaveGuard, an IoT coastal hazard monitoring system for real-time Kallakkadal detection along the Kerala coastline.
- Achieved Elite status in NPTEL Programming in Java (IIT Kharagpur), a Ministry of Education funded national certification.
- Presented technical projects at internal and external college-level reviews and viva examinations.